- Project Title: Smart Employee Attendance, Analytics, and Leave Management System

- 1. Project Objective

Build a complete, secure, responsive, and production-ready Employee Attendance, Analytics, and Leave Management System for my startup.

The application must allow employees to record daily attendance through secure individual access while enabling the administrator to manage employees, monitor attendance in real time, review leave requests, analyze workforce attendance patterns, calculate working hours, and download detailed reports.

The application must have three connected interfaces:

1. Employee Attendance Portal: A reusable, secure attendance form for check-in, check-out, and leave submission.

2. Admin Analytics Dashboard: A visually appealing, interactive dashboard providing real-time workforce information, analytical charts, KPIs, trends, filters, and administrative controls.

3. Individual Employee Dashboards: A dedicated dashboard for every employee, displaying that employee's attendance history, leave information, working hours, and performance-related attendance statistics.

- Mandatory dashboard access policy

In the initial release, all individual employee dashboards must be accessible only to authorized administrators.

Employees may use their authorized attendance portal, but they must not access their individual dashboards until I explicitly instruct you to enable that functionality in a future update.

Prepare the architecture for future employee dashboard access without enabling it prematurely. When employee dashboard access is eventually enabled, each employee must be restricted to their own information.

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- 2. Employee Registration and Authentication

Implement secure individual employee identities.

Each employee profile must include:

- Unique Employee ID
- Employee Name
- Work Email
- Department
- Designation
- Date of Joining
- Employment Status
- Account Status
- Assigned Work Schedule
- Weekly Off Days
- Authentication Credentials or Secure Individual Attendance Link

Prefer individual employee login accounts. Alternatively, provide secure employee-specific attendance links with appropriate identity verification, token security, and revocation controls.

Requirements:

- The employee's identity must be determined by the authenticated account or verified individual token.

- Employees must not be able to select another employee's identity or submit attendance on their behalf.

- Employees must not be able to change their Employee ID through the browser or API.

- Administrators must be able to register, edit, deactivate, and manage employee accounts.

- Deactivating an employee must prevent new attendance submissions without deleting historical records.

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- 3. Employee Attendance Portal

Create a modern, mobile-friendly attendance form accessible through a reusable, authorized employee-specific link or secure login.

The same link must work every day without requiring a new link each morning.

Automatically populate the employee's name and Employee ID from their authenticated identity.

- 3.1 Attendance fields

Display:

- Employee Name
- Employee ID
- Attendance Date, generated using the company timezone
- Attendance Type:
  - Present
  - Absent
- 3.2 Present workflow

When Present is selected, show:

- Full Day
- Half Day

The first valid submission for the employee and attendance date must create the official daily attendance record and record the check-in timestamp using the server.

When the employee accesses the same link again on the same day, retrieve the existing record and display the recorded check-in time.

If the employee has not checked out, display a Check Out action. On confirmation, record the check-out timestamp and update the same daily attendance record.

The database must maintain one official daily attendance record per employee per attendance date. Check-in and check-out must be stored in the same record.

Do not create a second attendance record when the employee checks out.

If the employee has already checked out, display the completed attendance information without overwriting the timestamps. Any necessary changes must go through an authorized administrative correction workflow.

Prevent duplicate submissions, simultaneous duplicate check-ins, and accidental timestamp overwrites.

- 3.3 Absent and leave workflow

When Absent is selected, display the following leave types:

- PL — Privilege Leave
- CL — Casual Leave
- Medical Leave

Require a supporting document in PDF, JPG, JPEG, or PNG format.

Validate the actual file format and file size on the server. Store documents securely and restrict their access to authorized personnel.

Allow a leave reason and, where applicable, the requested leave start and end dates.

Every submitted leave request must initially have Pending status.

An employee selecting a leave type and attaching a document must not automatically be considered on approved leave. The administrator must explicitly approve or reject the request.

Maintain separate attendance records and leave request records, linked through employee IDs and relevant dates.

Do not generate normal check-in or check-out timestamps for an absent employee.

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- 4. Automatic Date and Time Recording

Use server-generated timestamps as the authoritative source.

Record the following, as applicable:

- Attendance Date
- Check-In Timestamp
- Check-Out Timestamp
- Submission Timestamp
- Record Creation Timestamp
- Last Modified Timestamp
- Leave Request Timestamp
- Leave Decision Timestamp

Default company timezone: Asia/Kolkata.

Store timestamps in UTC where appropriate and display them using the configured company timezone.

Implement:

- Correct attendance date assignment near midnight
- Support for overnight shifts
- Duplicate request protection
- Preservation of original timestamps
- Audit logging for administrative corrections
- Missing check-out detection

Never rely on an employee's device clock as the authoritative timestamp. Never invent a missing check-out time.

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- 5. Admin Analytics Dashboard — High Priority

Build a visually attractive, analytical, interactive, and professional Admin Dashboard suitable for a modern startup.

The dashboard must do more than display basic attendance numbers. It should help the administrator understand workforce attendance patterns, identify irregularities, monitor leave activity, compare employees and departments, and make decisions using reliable data.

Use a clean enterprise-style visual design with a consistent color system, attractive data visualizations, clear typography, well-organized layouts, and intuitive navigation.

- 5.1 Visual design requirements

Use a professional interface with:

- A collapsible sidebar
- A clean top navigation bar
- A date-range and reporting-period selector
- Search and filter controls
- KPI summary cards
- Interactive charts and graphs
- Organized analytical sections
- A detailed attendance table
- Employee profile shortcuts
- Responsive layouts for desktop, tablet, and mobile
- Clear loading, error, empty, and success states
- Accessible colors, labels, and typography
- Tooltips and legends for data visualizations

Use a consistent design system rather than displaying unrelated charts and cards without visual hierarchy.

Use an appropriate charting library such as Recharts or an equivalent library supported by the chosen frontend stack.

Avoid excessive decoration, unnecessary animations, and overcrowded layouts. Prioritize readability and actionable information.

- 5.2 Top-level KPI cards

Display dynamically calculated KPI cards for the selected date or reporting period.

Include:

1. Total Active Employees
2. Present Employees
3. Full-Day Attendance
4. Half-Day Attendance
5. Absent Employees
6. Employees Currently Checked In
7. Employees Who Have Checked Out
8. Pending Leave Requests
9. Approved Leave Requests
10. Rejected Leave Requests

11. Missing Check-Outs

12. Late Arrivals

Where applicable, also display:

- Attendance Rate
- Absence Rate
- Average Recorded Working Hours
- Total Recorded Working Hours
- Employees with Incomplete Attendance Records

Each KPI must have a clear label and a meaningful value. Where a valid comparison period exists, show the change relative to the previous equivalent period.

Do not invent percentage changes or compare incompatible reporting periods.

All metrics must be calculated from actual database records. No hardcoded figures or simulated production data.

- 5.3 Attendance analytics charts

Implement the following interactive analytical charts.

A. Daily Attendance Distribution

Show the number of employees in each applicable attendance category:

- Present
- Half Day
- Absent
- Approved Leave
- Pending Leave

Use clearly distinguishable categories and ensure pending requests are not confused with approved leave.

B. Attendance Trend

Create a daily or weekly line chart showing attendance trends over the selected reporting period.

Allow administrators to switch between:

- Daily
- Weekly
- Monthly

Show the appropriate attendance counts and rates for the selected interval.

C. Present vs. Absent Trend

Display a time-series chart comparing present and absent employee counts over the selected period.

D. Department-Wise Attendance

Display attendance comparisons across departments, including employee counts and relevant attendance rates.

Allow the administrator to select a department and inspect its records.

E. Leave Distribution

Visualize leave requests by type:

- PL
- CL
- Medical Leave

Allow the administrator to distinguish Pending, Approved, and Rejected requests.

F. Working-Hours Analytics

Display total and average recorded working hours across the selected period.

Where meaningful, compare employees or departments using validated working-hour records. Clearly identify incomplete records that cannot yet contribute to a complete-duration calculation.

G. Late-Arrival Analytics

Display late-arrival counts and trends for employees whose work schedules and late-arrival rules are configured.

H. Attendance Exception Analysis

Highlight operational exceptions, including:

- Missing check-outs
- Missing check-ins
- Duplicate submission attempts
- Attendance requiring administrative review
- Unresolved attendance records

I. Employee Attendance Comparison

Provide a ranked or sortable employee table showing relevant metrics such as recorded attendance days, absence counts, late arrivals, and recorded working hours.

Use neutral, descriptive labels. Do not label employees as poor performers based solely on incomplete or missing data.

- 5.4 Interactive chart behavior

Every chart must be connected to real application data.

Where appropriate, allow administrators to:

- Hover over chart elements to view exact values
- Change the reporting period

- Apply employee and department filters
- Select a chart category to inspect the underlying records
- Navigate from a chart to the relevant filtered attendance table
- Compare current and previous periods
- Export the filtered underlying records

Ensure that chart values, KPI cards, tables, and exports remain consistent with the same filters and reporting rules.

Use charts that suit the data rather than selecting chart types solely for visual appeal.

For example:

- Line charts for attendance trends
- Bar charts for department comparisons
- Donut charts for a small number of attendance categories
- Tables for detailed record inspection
- 5.5 Date-wise attendance table

Display:

- Employee ID

- Employee Name
- Department
- Attendance Date
- Attendance Status
- Full Day or Half Day
- Check-In Time
- Check-Out Time
- Recorded Working Hours
- Leave Type
- Leave Approval Status
- Submission Timestamp
- Authorized Attachment Access
- Administrative Actions

Provide search, sorting, pagination, date filters, department filters, status filters, and missing check-out filters.

Include a View Dashboard action for each employee, allowing the administrator to open that employee's dedicated dashboard.

- 5.6 Real-time dashboard updates

Update the dashboard when an employee:

- Checks in
- Checks out
- Submits a leave request

Also update it when an authorized administrator approves a leave request or corrects an attendance record.

Use WebSockets, Server-Sent Events, or an appropriate polling mechanism. Provide a manual refresh option as a fallback.

Refresh affected KPIs, charts, and tables without displaying stale or inconsistent data.

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- 6. Individual Employee Dashboards

Create a dedicated dashboard for every registered employee.

The administrator must be able to open any employee's dashboard through the employee management page or attendance table.

Each individual dashboard must contain the selected employee's:

- Employee profile
- Employee Name

- Employee ID
- Department
- Designation
- Date of Joining
- Employment Status
- Account Status
- Work Schedule
- Individual KPI cards
- Recorded Present Days
- Full Days
- Half Days
- Absence Records
- PL Requests
- CL Requests
- Medical Leave Requests

- Approved Leave Requests
- Pending Leave Requests
- Rejected Leave Requests
- Total Recorded Working Hours
- Average Daily Working Hours
- Late Arrivals
- Missing Check-Outs

Show the selected reporting period and allow custom date ranges.

- Attendance history

Provide a date-wise table containing attendance status, check-in, check-out, working duration, leave type, approval status, and administrative remarks.

- Attendance calendar

Provide a monthly calendar displaying attendance categories, leave statuses, holidays, weekly off days, and incomplete records using distinguishable indicators.

- Individual analytics

Provide employee-specific charts for:

- Attendance trends

- Recorded working hours
- Late-arrival trends
- Leave distribution
- Attendance exceptions

Do not calculate or display misleading metrics when there is insufficient data.

- Leave history

Show all leave requests, requested dates, submission times, approval statuses, decision history, and authorized access to supporting documents.

- Access policy

In the initial version, individual employee dashboards must be visible only to authorized administrators.

Employees must not be able to access these dashboards, their URLs, their API endpoints, or the corresponding data.

Wait for my explicit instruction before enabling employee dashboard access. If authorized later, each employee must see only their own permitted information, while administrator privileges remain restricted to authorized administrators.

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- 7. Employee Record Downloads — Excel and CSV

Implement a complete, reliable export and download system inside the Admin Dashboard.

The administrator must be able to download an individual employee's records as well as records for all employees in Excel or CSV format.

This functionality is mandatory.

- 7.1 Individual employee downloads

Inside every individual employee dashboard, provide a clearly visible Export / Download Records action.

Allow the administrator to download the selected employee's records as:

- Excel Workbook (.xlsx)
- Comma-Separated Values (.csv)

Allow the administrator to select:

- Start Date
- End Date
- Attendance Records
- Leave Records
- Working-Hours Records
- Attendance Exceptions
- Available administrative remarks

Include only the selected employee's records for the specified filters.

- 7.2 Company-wide bulk downloads

Inside the main Admin Dashboard and Reports section, provide a dedicated Export Reports option.

The administrator must be able to download records for:

- All active employees
- All employees, including inactive employees' historical records
- A selected employee
- Multiple selected employees
- A selected department
- A selected date range

Allow export filters for attendance status, leave type, leave approval status, department, and other applicable criteria.

- 7.3 Excel workbook structure

Generate a well-formatted Excel workbook with meaningful worksheet names.

For a company-wide export, include separate worksheets where appropriate, such as:

1. Attendance Records

2. Employee Directory

3. Leave Requests

4. Working Hours

5. Attendance Summary

6. Exceptions

For an individual employee export, include:

1. Employee Profile

2. Attendance History

3. Leave History

4. Working Hours

5. Attendance Summary

6. Exceptions

Each worksheet must include meaningful column headings, correct data types, readable formatting, and appropriate date/time formatting.

Avoid duplicating the same record across worksheets unnecessarily.

Include an export generation timestamp and the selected date range where applicable.

- 7.4 CSV export

Provide CSV downloads for:

- Attendance records
- Leave records
- Employee-specific records
- Company-wide filtered records

Each CSV must represent one clearly defined dataset with a header row.

Document the date format and time interpretation. Ensure fields containing commas, quotation marks, or line breaks are escaped correctly.

Protect exports against spreadsheet formula injection when fields contain untrusted text.

- 7.5 Export accuracy

All reports must be generated from actual database records and respect the active filters and the administrator's permissions.

The export logic must ensure:

- Correct employee IDs and names
- Correct attendance dates
- Accurate check-in and check-out timestamps

- Accurate working-hour calculations
- Correct Full-Day and Half-Day classifications
- Correct leave types and approval statuses
- No unintended duplicate rows
- Consistency between dashboard metrics and exported summaries

Pending leave must remain distinguishable from approved leave.

Do not fabricate missing timestamps or calculated values.

If working hours are unavailable because a check-out is missing, mark the record as incomplete rather than inventing a value.

- 7.6 Secure export handling
- Only authorized administrators can generate company-wide exports.
- Employees must not be able to download company-wide records.
- Individual employee exports must include only the intended employee's records.
- Medical documents must not be embedded in exports by default.
- Include document references or secure attachment metadata only where appropriate.
- Do not expose private file-storage paths, credentials, or authentication tokens.

- Generate exports securely and use temporary download files or an equivalent safe delivery mechanism.

- Clean up temporary generated files according to the configured retention policy.

- 7.7 Export user experience

When the administrator clicks Export:

1. Validate the selected filters.

2. Generate the requested report.

3. Display an appropriate loading state.

4. Provide the resulting file for download.

5. Display a useful error if generation fails.

For large datasets, use a background job or streamed export where appropriate, with clear status feedback.

Include automated tests for individual exports, company-wide exports, date filtering, row counts, correct worksheet contents, and access control.

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- 8. Leave Management and Approval

Create a dedicated leave management section.

Administrators must be able to:

- View submitted requests
- Review supporting documents
- Approve or reject requests
- Record decision timestamps
- Add administrative remarks
- Enter rejection reasons
- Inspect previous requests
- Filter by employee, date, department, leave type, and status

Every leave request must begin with Pending status.

Keep leave requests separate from attendance records. Update the corresponding attendance classification according to the company's configured policy when leave is approved or rejected.

A rejected request must remain in history, and its attendance consequences must not be changed silently.

If leave balances are implemented, make the entitlement and deduction rules configurable rather than assuming universal company policy.

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- 9. Working-Hours Calculations

Calculate recorded working duration from valid timestamps:

- \text{Check-Out Timestamp}

\text{Check-In Timestamp}

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Support:

- Configurable shifts
- Working-hour thresholds
- Late-arrival detection
- Early-departure detection
- Missing check-in and check-out alerts
- Overtime calculations where configured
- Full-Day and Half-Day classifications
- Authorized administrative corrections

Full Day and Half Day must be attendance classifications rather than assumptions about actual working hours.

Do not automatically invent a check-out time. Flag incomplete records for administrative review.

Handle overnight shifts through explicit business rules.

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- 10. Database Architecture

Use a normalized relational database with suitable constraints, foreign keys, and indexes.

Consider these tables:

1. users
2. employees
3. attendance_records
4. leave_requests
5. attendance_attachments
6. work_schedules
7. holidays
8. notifications
9. audit_logs

Apply a unique constraint on the combination of Employee ID and Attendance Date in attendance_records.

Maintain these principles:

- One official daily attendance record per employee and date
- Check-in and check-out update the same record
- Leave requests are stored separately
- Each employee dashboard loads the selected employee's records
- Dashboard and export endpoints enforce authorization
- Database transactions prevent concurrent duplicate records
- Historical records remain available after employee deactivation
- Administrative corrections are auditable

Use database aggregation queries or an efficient reporting service for dashboard KPIs and analytical charts. Avoid repeatedly loading the entire database into the frontend to compute summary metrics.

- 10.1 Detailed Database Design

Create and document the actual relational schema before implementing the application. Define each table's purpose, columns, data types, primary keys, foreign keys, relationships, indexes, uniqueness rules, nullability, and allowed status values. Use database migrations so schema changes are versioned and reproducible.

At minimum, implement the following design requirements:

- users: user ID, authentication identifier, password hash or identity-provider reference, role, account status, and creation/modification timestamps. Never store plaintext passwords or authentication secrets.

- employees: employee ID, linked user ID where applicable, employee number, name, work email, department, designation, date of joining, employment status, assigned schedule, and account status.

- attendance_records: record ID, employee ID, attendance date, attendance classification/status, check-in timestamp, check-out timestamp, recorded working duration where calculated, schedule reference, related leave reference where applicable, and creation/modification timestamps.

- leave_requests: request ID, employee ID, leave type, requested start/end dates, reason, supporting-document reference, request status, reviewer ID, review timestamp, decision remarks, and creation/modification timestamps.

- attendance_attachments: attachment ID, related employee and request/record IDs, original filename, validated media type, file size, private storage key, upload timestamp, and uploader ID. Do not store a publicly accessible file URL as the authorization mechanism.

- work_schedules: schedule ID, schedule name, expected start/end times, working days, applicable break rules, late-arrival threshold, and effective dates.

- holidays: date, description, scope/department if applicable, and configured holiday type.

- notifications: recipient, event type, related record, delivery/read status, and timestamps.

- audit_logs: actor, action, entity type and ID, timestamp, reason where required, and a protected record of relevant before/after values. Do not allow ordinary users to edit or delete audit entries.

Use stable internal IDs for relationships; do not use employee names as keys. An employee number and a login identity must be unique where required. Enforce a database-level unique constraint on (employee_id, attendance_date) in attendance_records. Use foreign keys and appropriate delete restrictions so deleting or deactivating an employee cannot silently erase attendance, leave, attachment, or audit history.

Store timestamps using timezone-aware database types, preferably in UTC, and store the official attendance date separately according to the configured company timezone and shift policy. Define allowed attendance and leave status values explicitly, using suitable enum types or constrained values. Keep the attendance status and leave request approval status as separate concepts.

Add indexes for common queries, including employee/date, date/status, department/employee, leave status/date, and any fields used frequently by dashboard filters. Avoid unnecessary indexes that slow down writes. Use pagination and efficient database queries for large record sets.

Use migrations to create and update the schema. Include a safe initial administrator setup process, development-only seed data where useful, database connection configuration through environment variables, backup/restore guidance, and a documented data-retention policy. Do not include real employee personal data in development fixtures.

- 10.2 Essential Business Logic and State Transitions

Implement core attendance and leave rules in the backend and database layer, not only in frontend conditions. Define and test valid state transitions before building dashboard interactions.

A. Daily attendance state machine

1\. Resolve the authenticated employee and the official attendance date using the configured timezone and shift policy.

2\. If no daily attendance record exists and the employee submits Present, create one record in a single atomic transaction, record the server-generated check-in timestamp, and save the selected Full Day or Half Day classification.

3\. If a record exists with a valid check-in and no check-out, the next valid check-out action must update that same record with the server-generated check-out timestamp. It must not insert another daily record.

4\. If both check-in and check-out already exist, treat attendance as completed. Do not overwrite either timestamp through an ordinary employee submission.

5\. If a leave request or absence-related status already exists for that date, do not reinterpret the next request as a check-out. Show the current state and follow the configured correction/review process.

6\. If the employee has already checked in and then attempts to submit Absent for the same date, do not silently replace the present record. Block the conflicting action and direct the employee to the approved correction process.

7\. Enforce one official attendance record per employee and attendance date through a database constraint, not just a frontend check.

B. Idempotency and concurrency

Use idempotency keys or equivalent request deduplication for attendance-changing operations. Handle rapid double-clicks, browser retries, network timeouts, and simultaneous requests safely. Use database transactions, row-level locking or equivalent concurrency controls, and uniqueness constraints to ensure that two concurrent submissions cannot create two check-ins or two daily records. Return the existing valid result when the same request is retried rather than creating another event.

C. Leave lifecycle

1\. Every new leave request starts as Pending.

2\. A request is not approved merely because an employee selected a leave type or attached a document.

3\. Only an authorized administrator can approve or reject a request.

4\. Store who reviewed the request, the decision, the decision timestamp, and remarks/rejection reason.

5\. Keep the original request and every decision/correction traceable. A rejected request must not be deleted or silently converted to approved leave.

6\. For a request spanning several dates, associate it with each applicable date while preserving one official attendance record per employee/date. Prevent duplicate daily records.

7\. Detect overlapping leave requests and leave requests that conflict with an existing check-in or finalized attendance record. Apply explicit company-policy rules and require review where needed rather than silently overriding data.

8\. When a request is approved or rejected, update the applicable daily attendance classification only according to the configured company policy. Pending leave must remain distinguishable from approved leave.

9\. If a leave request is still pending for a date that has passed, keep the request pending until reviewed; do not present it as approved leave. If rejected, follow the configured process for final attendance classification and retain the decision history.

D. Working time and attendance classification

Calculate recorded working duration only when valid check-in and check-out timestamps are available. Validate that the checkout is later than the check-in according to the assigned shift and overnight-shift rules. Do not infer a check-out time from the shift end or from a later activity. Treat Full Day/Half Day as attendance classifications; determine lateness, early departure, and overtime from configured schedules and rules, not from the classification alone.

E. Date, shift, and holiday rules

Determine attendance dates using the configured company timezone and the employee's assigned shift. Define how a shift crossing midnight is assigned to its attendance date. Apply weekly off days and holidays consistently. Do not automatically mark a holiday or weekly off day as an absence unless the company's configured policy explicitly requires it. Use a documented cut-off or administrator review process before finalizing missing attendance as absent.

F. Administrative corrections

Only authorized administrators may correct finalized attendance data. Require a reason where appropriate, preserve the original value, record the new value and the actor/time in the audit log, and do not permit direct database overwrites that bypass the audit trail. Corrections must update dashboard aggregates and exports consistently.

G. Analytics and reporting rules

Centralize definitions for metrics such as attendance rate, absence rate, present days, leave counts, average working hours, late arrivals, and missing check-outs. Document the numerator, denominator, exclusions, and applicable date range for each metric. Exclude records that cannot support a valid calculation and label incomplete data clearly. Ensure KPI cards, charts, tables, individual dashboards, and downloadable reports use the same filters and calculation rules.

H. Export rules

Generate exports from authorized server-side queries using the selected filters. Use a consistent inclusive/exclusive date-boundary convention and display the applied date range and timezone. Prevent duplicate rows caused by joins between attendance and multi-day leave requests. Treat attendance and leave as distinct datasets where appropriate, and protect CSV/XLSX output against spreadsheet formula injection from untrusted text.

- 10.3 Edge Cases and Failure Handling

Explicitly handle and test the following cases without losing data, inventing timestamps, bypassing permissions, or creating inconsistent records:

1\. An employee submits check-in twice, checks out twice, double-clicks the action, refreshes the page, or retries after a timeout.

2\. Two or more requests arrive simultaneously for the same employee and attendance date.

3\. Two employees have the same name, or an employee changes a display name after historical records exist.

4\. An employee attempts to change an Employee ID, use another employee's link, reuse an expired/revoked link, or access another employee's dashboard or records.

5\. An employee account is deactivated while a session or attendance link is still open.

6\. The employee checks in just before midnight and checks out after midnight, including an overnight shift assigned to a prior work date.

7\. A submission arrives near a company-local date boundary, or the configured timezone changes. Use the documented timezone and shift rule consistently.

8\. Check-out is attempted before check-in, is earlier than check-in, or cannot be matched to an allowed overnight shift.

9\. An employee forgets to check out, checks in but never submits again, or has only one of the two timestamps. Mark the record incomplete and flag it for review; do not fabricate a timestamp.

10\. An employee attempts to submit Absent or leave after a check-in already exists, or attempts Present after a leave request has been submitted.

11\. A leave request is pending, approved, or rejected after the requested date has passed.

12\. Two leave requests overlap, a request covers multiple dates, a leave date falls on a holiday/weekly off day, or a request conflicts with an existing attendance record.

13\. A leave request is rejected and the company needs a final attendance classification. Retain the original request and apply the configured attendance policy with an audit trail.

14\. An employee submits a request on a holiday, weekly off day, outside a configured shift, or when the employee is inactive.

15\. A supporting document has an unsupported extension, incorrect file signature, excessive size, suspicious content, an unsafe filename, or duplicate upload.

16\. File storage succeeds but the database transaction fails, or the database transaction succeeds but file storage fails. Handle partial failure safely and clean up orphaned files/records as appropriate.

17\. A request loses network connectivity or the real-time update channel disconnects. Persist the authoritative operation once and recover the dashboard using a safe refresh/retry strategy.

18\. A database outage, deadlock, validation failure, or unexpected server error occurs during check-in, check-out, leave submission, approval, or export generation.

19\. A dashboard filter returns no rows, an employee has insufficient historical data for a metric, or a date range includes incomplete records. Show a clear empty or incomplete state instead of misleading analytics.

20\. A bulk export contains a large number of records, a field contains commas/newlines/quotes, or untrusted text begins with a spreadsheet formula character. Export safely without exposing data beyond the administrator's permissions.

21\. An export or attachment download URL is shared, expired, or requested by an unauthorized user. Recheck authorization server-side and use protected or short-lived access.

22\. An administrator corrects attendance after a report has been generated. Future reports and dashboard aggregates must reflect the correction, while the audit history preserves the original values.

23\. A leave request or attendance record is associated with an inactive employee. Preserve historical access for authorized administrators without allowing new submissions from the inactive account.

24\. A scheduled shift, leave policy, holiday calendar, or employee department changes over time. Preserve the historical context required to interpret prior attendance accurately.

25\. A failed or repeated approval request arrives, or two administrators attempt to decide the same leave request simultaneously. Prevent conflicting decisions and record the authoritative outcome.

For every failure case, provide a clear user-facing message, log appropriate diagnostic details without exposing secrets or medical information, preserve data integrity, and avoid revealing whether protected records exist to unauthorized users. Define retry behavior and administrative recovery steps where needed.

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- 11. Security and Access Control

Implement:

- Secure authentication
- Role-based authorization
- Server-side validation
- Secure password hashing
- Protection against common web vulnerabilities
- CSRF protection where applicable
- File-upload validation
- Private document storage
- Secure export handling
- Audit logging
- Database transactions
- Idempotent attendance requests
- Employee identity verification
- Revocation of compromised attendance links
- Environment-based configuration for secrets

At minimum, support the following permissions.

The application must implement proper authentication, authorization, and data protection.

- Only authorized administrators may access the admin dashboard.

- Employees must only be able to view their own records unless explicitly authorized otherwise.

- Prevent employees from editing timestamps or changing another employee's attendance.

- Validate all inputs on both client and server.

- Validate uploaded files using file signatures and permitted formats, not just filenames.

- Restrict file sizes and store uploaded documents in private storage.

- Use authenticated, time-limited access to sensitive attachments.

- Protect against common web vulnerabilities, including SQL injection, cross-site scripting, cross-site request forgery where applicable, and broken access control.

- Store passwords securely using an established password-hashing algorithm.

- Use environment variables for credentials and secrets.

- Maintain an audit log for administrative changes.

- Apply database constraints to prevent duplicate attendance records for the same employee and attendance date.

- Use transactions or atomic operations so simultaneous requests cannot create duplicate check-ins.

The employee attendance link must not expose sensitive information or allow an unauthenticated visitor to impersonate any employee.

Administrator: Can access the admin analytics dashboard, every individual employee dashboard, authorized employee records, leave review functionality, attendance corrections, company reports, and company-wide exports.

Employee: In the initial version, can access only the authorized attendance portal and applicable personal submission functions. Individual employee dashboard access and record-history dashboards must remain disabled until explicitly authorized by me.

If employee dashboard access is enabled later, every request must be restricted to the authenticated employee's permitted data.

Do not rely only on hiding dashboard navigation items. Protect routes, APIs, records, documents, and downloads on the backend.

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- 12. Recommended Technology Stack

Use the following technology stack unless there is a clear technical reason to choose an equivalent alternative:

- Frontend: React with TypeScript

- UI: Tailwind CSS and reusable accessible components

- Charts: Recharts or an equivalent visualization library

- Backend: Python with FastAPI
- Database: PostgreSQL
- ORM and Migrations: SQLAlchemy and Alembic
- Authentication: Secure session-based or appropriately configured token-based authentication
- File Storage: Private object storage or secure server-side storage
- Real-Time Updates: WebSockets or Server-Sent Events, with polling as a fallback
- Analytics: Efficient database aggregations and validated business logic
- Excel Export: Pandas with a suitable Excel workbook generation library
- CSV Export: Secure, correctly escaped CSV generation
- Deployment: Docker-compatible production configuration

Use a modular architecture so the attendance portal, main admin dashboard, individual dashboards, analytics, reporting, export service, and leave workflow can be maintained independently.

- 13. Local Development and Execution — Mandatory Initial Setup

For the initial version, develop, configure, and run the entire Employee Attendance, Analytics, and Leave Management System locally on my Windows computer. Do not deploy it to the cloud at this stage.

Technology and local services

- Frontend: React with TypeScript.

- Backend: Python with FastAPI.

- Database: PostgreSQL installed and running locally on Windows.

- ORM and migrations: SQLAlchemy and Alembic.

- Frontend package manager: npm.

- Python dependencies: Use a project-specific virtual environment.

- File storage: A local private directory managed by the backend, outside publicly accessible frontend assets.

Setup instructions

1. Inspect my computer's available development tools and check which prerequisites are already installed.

2. Provide installation instructions only for missing prerequisites.

3. Create the frontend and backend project structures.

4. Generate the dependency files and install the necessary packages.

5. Configure a local PostgreSQL database and explain how to create the database and its credentials.

6. Store configuration in a local `.env` file, using a `.env.example` template for documentation. Never commit passwords or secrets to source control.

7. Configure SQLAlchemy and Alembic, create the migrations, and initialize the database.

8. Implement the complete backend, frontend, authentication, attendance workflows, leave management, analytics, individual employee dashboards, and Excel/CSV exports.

9. Store uploaded documents in a protected local directory. Do not expose medical documents or attachments as publicly accessible static files.

10. Create a `.gitignore` that excludes secrets, virtual environments, generated files, private uploads, and other sensitive local data.

11. Provide exact Windows PowerShell commands to start the backend and frontend in separate terminals.

12. Verify that attendance submissions are persisted in PostgreSQL and remain available after restarting the application.

13. Provide a local database backup and restore procedure.

14. Test the application locally, fix errors, and document the actual results.

Runtime requirements

- Frontend and backend must run on localhost.

- The backend must connect to the local PostgreSQL instance.

- No cloud database, cloud file storage, external hosting, paid subscription, or public deployment should be required for local operation.

- Do not create paid services or publish employee information.

- Provide separate instructions for stopping and restarting the frontend, backend, and database services.

- Ensure the application can be started again after restarting Windows without losing saved employee records.

- Do not claim that the application is running until its services and database connection have been verified.

- Keep employee dashboard access disabled in the initial version, as specified in the main project requirements.

Complete and verify local operation before proposing any future cloud deployment.

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- 14. UI Navigation

Create a sidebar with the following sections:

- Overview

- Attendance Management

- Employee Management

- Individual Employee Dashboards

- Leave Management

- Attendance Calendar
- Analytics and Insights
- Reports and Downloads
- Work Schedules
- Notifications
- Audit Logs
- System Settings

The Overview page must show the key analytics and high-priority attendance exceptions.

The Analytics and Insights page must contain the more detailed interactive charts and workforce trends.

The Reports and Downloads page must let administrators generate and download filtered Excel and CSV reports.

Provide a View Dashboard action for every employee in the employee directory and attendance table.

Make the interface fully responsive and maintain a consistent professional design across all sections.

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- 15. Automated Testing and Edge Cases

Test at least the following cases:

1. Initial check-in and subsequent check-out.
2. New attendance record on the following day.
3. Duplicate submissions and simultaneous requests.
4. Employees with identical names.
5. Attempts to impersonate another employee.
6. Leave submission with a valid attachment.
7. Pending, approved, and rejected leave requests.
8. Invalid or oversized attachments.
9. Missing check-out.
10. Overnight shifts and timezone boundaries.
11. Network failures and repeated requests.
12. Administrative attendance corrections.
13. Deactivated accounts and revoked links.
14. Unauthorized access to individual dashboards.
15. Administrator access to every authorized employee dashboard.

16. Employee dashboard access remaining disabled initially.

17. Correct employee-only data isolation if later explicitly enabled.

18. Correct KPI calculations and chart filters.

19. Individual employee Excel and CSV downloads.

20. Company-wide Excel and CSV downloads.

21. Date-range and department filtering in exports.

22. Correct worksheet structure and export row counts.

23. Consistency between displayed KPIs and exported totals.

24. Unauthorized download attempts.

25. Export handling for missing timestamps and rejected leave.

26. Database or file-storage failures.

Run automated tests for critical business rules and fix identified defects before considering the system complete.

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- 16. Development Phases

Phase 1 — Planning and Architecture: Define the architecture, database schema, access rules, and attendance state transitions.

Phase 2 — Employee Registration and Authentication: Implement employee profiles, secure authentication, and individual attendance access.

Phase 3 — Attendance Workflow: Implement check-in, check-out, date-wise records, absence submission, and server timestamps.

Phase 4 — Admin Dashboard: Build the visually polished, interactive Admin Dashboard with KPIs, analytical charts, filters, attendance tables, and employee management.

Phase 5 — Individual Employee Dashboards: Implement dedicated dashboards for every employee, with access restricted to authorized administrators in the initial version.

Phase 6 — Leave Management: Implement document uploads, leave review, approval, rejection, and audit history.

Phase 7 — Analytics and Reports: Implement attendance trends, department analysis, working-hours analytics, attendance exceptions, and report generation.

Phase 8 — Excel and CSV Downloads: Implement individual employee exports, bulk employee exports, filtering, secure generation, and download functionality.

Phase 9 — Security and Testing: Validate authorization, database constraints, uploads, analytics, exports, and edge cases.

Phase 10 — Deployment: Prepare configuration, migrations, backup and recovery documentation, and deployment instructions.

Do not enable individual employee dashboard access unless I explicitly instruct you to do so.

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- 17. Project Deliverables

Deliver:

- Complete frontend and backend source code
- Database schema and migrations
- Working employee attendance portal
- Secure administrator authentication
- Visually polished Admin Dashboard
- Interactive attendance analytics and charts
- Dedicated individual employee dashboards
- Leave submission and approval workflow
- Excel and CSV downloads for individual employees
- Excel and CSV downloads for all or filtered employees
- .env.example with placeholder values
- Automated tests
- README.md with setup and execution instructions
- Secure first-administrator creation instructions
- Database backup and restore documentation

- Deployment guide

- Test results and a list of remaining limitations

Do not use hardcoded attendance statistics or mock production records. Mark features as completed only after implementation and testing.

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- 18. Final Acceptance Criteria

The project is complete only when:

- Employees can securely submit attendance.

- Check-in and check-out timestamps are recorded automatically.

- Both timestamps are maintained in the same daily attendance record.

- Duplicate attendance records and timestamp overwrites are prevented.

- Leave submissions require supporting documents.

- Leave requests require an explicit administrative decision.

- The administrator can monitor attendance in real time.

- The Admin Dashboard is visually polished, analytical, and responsive.

- Dashboard cards, charts, tables, and exports use actual database data.

- Analytics can be filtered by reporting period, department, employee, and applicable attendance or leave status.

- The administrator can open every employee's individual dashboard.

- Individual dashboards display attendance, working hours, leave history, and employee-specific analytics.

- Individual dashboard access remains restricted to administrators in the initial version.

- The administrator can download one employee's attendance and related records as XLSX or CSV.

- The administrator can download company-wide or filtered employee records as XLSX or CSV.

- Exports respect filters, match database records, and protect sensitive information.

- Dashboard totals and export totals are consistent.

- Unauthorized users cannot access dashboards, records, attachments, or exports.

- Tests verify attendance logic, analytics calculations, and export accuracy.

- Setup, testing, and deployment documentation are complete.

- 19. Instructions to the AI Coding Agent

First, inspect the existing project workspace and explain the proposed architecture, database schema, attendance workflow, and access-control approach.

Then implement the application systematically, beginning with secure authentication, database design, and attendance business logic.

Build the Admin Dashboard as a first-class product feature, not as a basic collection of cards and tables. It must be visually professional, analytically useful, interactive, and connected to real data.

Implement individual employee dashboards and administrator-only access. Add robust Excel and CSV downloads that support individual employees, multiple selected employees, departments, date ranges, and company-wide reporting.

Do not use mock figures, fabricate timestamps, or implement frontend-only security. Integrate all components with the real backend and database.

Run the available automated tests, resolve errors, and verify attendance submissions, leave approval, dashboard analytics, employee data isolation, and Excel/CSV exports.

Finally, provide a summary of the implemented features, application structure, local setup instructions, test results, and any remaining limitations.

Important: Individual employee dashboards must remain accessible only to the administrator in the initial release. Wait for my explicit instruction before enabling employee dashboard access.